

Program Relocation

- Absolute Program : Program is loaded at specific memory which is specified at assembly time

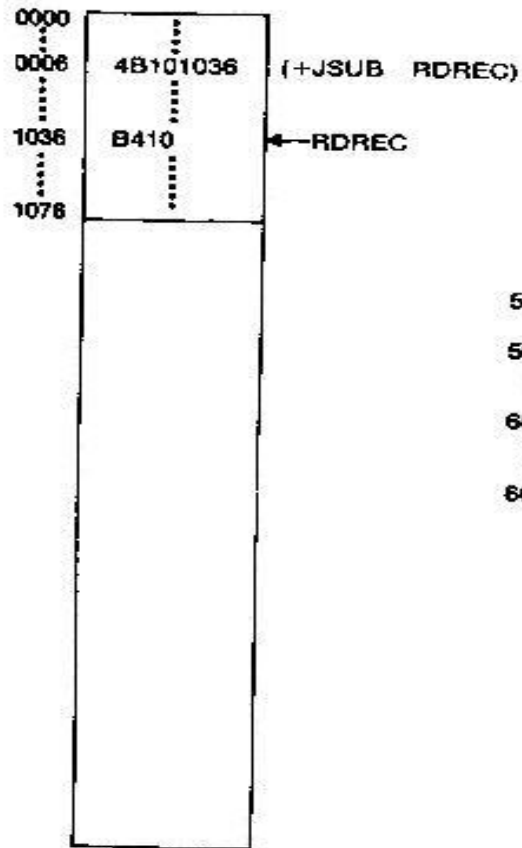
Eg: Starting address 1000

- In real time systems, where many simultaneous programs are executed, assembler will not know where the program is actually loaded.
- Parts of the addresses in object program need to be modified accordingly based on where program is actually loaded

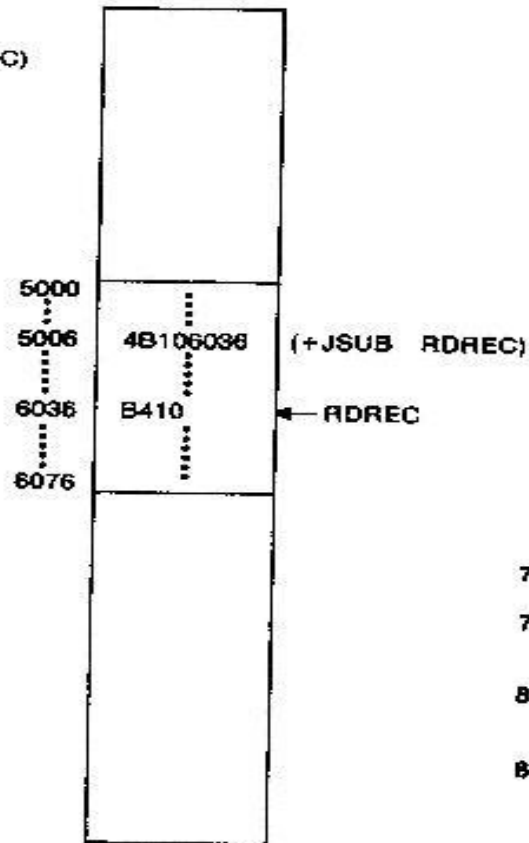
Program Relocation

- Therefore, assembler generates **Relocatable Program**. The object program contains information regarding the parts of the program that need modification. The object program generated by assembler starts at an address 000000.

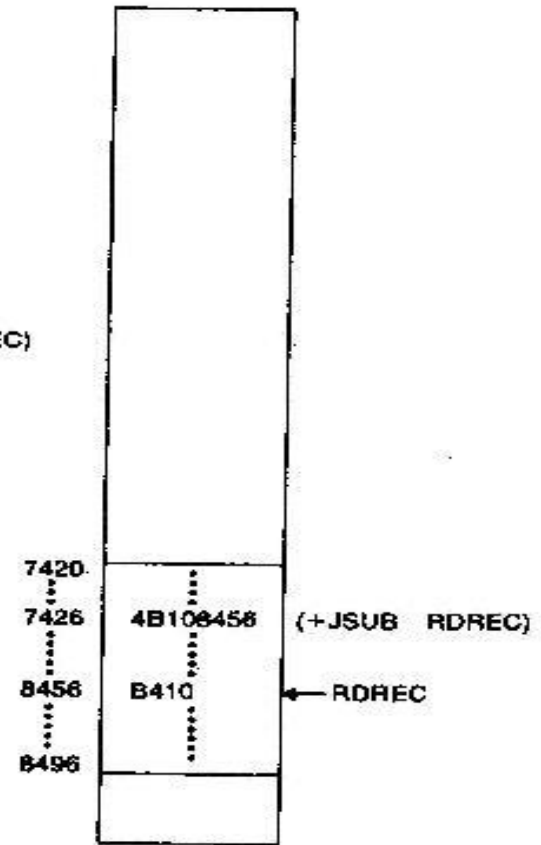
Example



(a)



(b)



(c)

Program Relocation

NOTE:

Object codes having following do not require any modification

- PC-relative and Base-relative addresses
- not a memory address (immediate addressing)

The only parts of the program that require modification at load time are those that specify direct addresses (Format-4 instructions in SIC/XE)

Program Relocation

2 methods for specifying relocation as part of object program:

- 1. Using Modification Records**
- 2. Using Bit Mask**



Relocation using Modification Records

Relocation using Modification Records

- Most of the instructions in SIC/XE use PC Relative or Base Relative addressing. Some use immediate addressing and register addressing. These instructions do not require any modification.
- Instructions that contain actual addresses are extended format (format-4). These require modification during relocation.

Relocation using Modification Records

▣ **Modification record Format**

- ▣ Col 1 M
- ▣ Col 2-7 Starting location of the address field to be modified, relative to the beginning of the program
- ▣ Col 8-9 length of the address field to be modified, in half-bytes

Relocation using Modification Records

Line	Location				Object Code
1	000000	START	COPY	O	
2	000000	FIRST	STL	RETADR	
3	000003		LDA	LENGTH	
4	000006		+JSUB	WRREC	4B101036
....					
....					
10	000026		+JSUB	RDREC	4B10105D
11		BUFFER	RESW	1	
					

Relocation using Modification Records

- In the above sample SIC/XE program, the instructions that are affected by relocation are at lines 4 and 10. They are extended format instructions

000006		+JSUB	WRREC	4B101036
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Modification is needed to last **05 half Bytes**.

i.e., 4B1**01036**

Address of the instruction = 000006

Modification is needed starting at address = **000007**

Relocation using Modification Records

000026		+JSUB	RDREC	4B10105D
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Modification is needed to last **05 half Bytes**.

i.e., 4B1**0105D**

Address of the instruction = 000026

Modification is needed starting at address = **000027**

Relocation using Modification Records

Object Program with Modification Records

Header Record.....

Text Record 1.....

Text Record 2.....

M_000007_05

M_000027_05

E_000000

Relocation using Modification Records

Advantage:

The Modification record scheme is a convenient way of specifying relocation

Disadvantage:

- ❑ As number of format-4 instructions increase, the number of modification records also increase. Thereby increasing the size of object program
- ❑ It is not recommended for a machine that uses direct addressing mode



Relocation using Bit Mask

Relocation using Bit Mask

- The text records in object program have a relocation bit associated with each word of object code
- It is suitable when large number of relocations are needed.
- It is suitable for machines where only direct addressing mode is used (Standard SIC)

Relocation using Bit Mask

Text record Format when Bit Mask is used

- Col 1 T
- Col 2-7 Starting address
- Col 8-9 length in Bytes
- **Col 10-12: Relocation Bits**
- Col 13-72: Object Codes

Relocation using Bit Mask

Example:

T_000000_1E_140003_4C0034_481061_....._00002D

If in the above text record, modification has to be done for first 3 object codes. Then the bit mask is 111000000000. Its hexadecimal value is E00.

The corresponding Text record with Bit Mask is

T_000000_1E_**E00**_140003_4C0034_481061_.....
_00002D

Relocation using Bit Mask

- A bit value 0 in mask represents no modification is necessary
- If relocation bit corresponding to an object code is set to 1, then it has to be modified. Starting address of the program is added to it when the program is relocated
- The relocation bits are gathered together into a Bit Mask which is represented as 3 Hexadecimal digits.
- If a text record contains fewer than 12 words of object codes , the bits corresponding to unused words are set to 0